Chinese Journal of Electronics
vol. 35, no. 3, pp. 1073–1086, May 2026
https://doi.org/10.23919/cje.2025.00.465

Signal Processing: Theory, Methods, and Applications

RESEARCH ARTICLE

Open-Set Recognition for Low-Altitude UAV Surveillance Using Radar Micro-Doppler Signatures

Guidong He[1], Peng Lei[1], Jun Wang[2], and Jiangyou Zhu[3]

1. *The School of Electronic and Information Engineering, Beihang University, Beijing 100191, China*
2. *Hangzhou Innovation Institute of Beihang University, Hangzhou 310000, China*
3. *The Chinese University of Hong Kong, Hong Kong 999077, China*

Corresponding author: Peng Lei, Email: buaaray@gmail.com.
Manuscript Received October 14, 2025; Accepted January 5, 2026; Published Online January 16, 2026
Copyright © 2026 Chinese Institute of Electronics

Abstract — Rotorcraft unmanned aerial vehicle (UAV) recognition is an crucial task in radar-based low-altitude UAV surveillance. Micro-Doppler signatures offer promising capabilities for UAV recognition. However, the diversity and unpredictability of other low-altitude targets pose significant challenges to conventional closed-set recognition methods. To address this issue, we propose an open-set recognition approach for identifying rotorcraft UAVs and other targets. First, a boundary point identification method based on the minimum description length (MDL) code length is introduced. Then, based on the MDL-induced code length, it is theoretically demonstrated that the extreme boundary samples conform to a generalized extreme value distribution, which enables the establishment of statistically sound decision thresholds. Finally, experiments conducted on real-world radar datasets demonstrate the superiority of our work over existing methods.

Keywords — Open-set recognition, Micro-Doppler, Unmanned aerial vehicle, Code length, Minimum description length, Generalized extreme value (GEV).

Citation — Guidong He, Peng Lei, Jun Wang, *et al.*, "Open-set recognition for low-altitude UAV surveillance using radar micro-Doppler signatures," *Chinese Journal of Electronics*, vol. 35, no. 3, pp. 1073–1086, 2026. doi: 10.23919/cje.2025.00.465.

I. Introduction

Unmanned aerial vehicle (UAV) monitoring has become increasingly critical in both civilian and defense applications, including airspace management, infrastructure surveillance, and security enforcement [1]. Among the various sensing modalities, radar stands out as a robust and all-weather solution capable of long-range target detection and velocity estimation [2].

Distinguishing rotorcraft UAV types is an important task in UAV monitoring, which is crucial for threat assessment, adaptive monitoring strategies, and enhancing the robustness of radar-based recognition systems [3]. However, other low-altitude targets, such as birds and fixed-wing UAVs, pose a significant challenge in this task. Due to their similar size, flight patterns, and radar cross-section (RCS) characteristics, their echoes often closely resemble those of rotorcraft UAVs [4], [5]. This similarity introduces considerable ambiguity in radar signal interpretation and significantly damages the performance of automatic recognition systems. As a result, traditional radar-based classifiers may struggle to distinguish between rotorcraft UAVs and these low-altitude targets [6], leading to a high rate of false alarms and reduced system reliability.

To tackle this challenge, micro-Doppler signatures [7] have been leveraged to distinguish rotorcraft UAVs from other targets. For example, birds generate micro-Doppler features primarily from wing flapping, resulting in periodic sidebands around the main Doppler frequency. These patterns are often asymmetric and vary with wingbeat rate and posture [8]. In contrast, rotorcraft UAVs, produce more stable and harmonic micro-Doppler features due to their rotating propellers [9].

To date, numerous studies have achieved promis-

Associate Editor: Prof. Xiaolong Chen, Naval Aviation University.

ing results by leveraging micro-Doppler signatures to classify rotorcraft UAVs and other low-altitude targets [10]–[13]. Many of these approaches, however, are based on closed-set recognition, which assumes that all classes appearing in the test set are already known and represented in the training set. In practice, low-altitude targets are highly diverse, inherently uncontrollable, and may appear unpredictably. As a result, the test set often contains classes beyond those seen during training. Under such conditions, closed-set methods tend to misclassify unseen targets into the closest known categories, leading to unreliable recognition.

These limitations highlight the necessity of open-set recognition (OSR) techniques [14], which are designed to not only classify known targets accurately but also identify and reject previously unseen classes. In this context, developing robust OSR methods tailored to radar-based UAV monitoring is of both theoretical interest and practical significance.

OSR methods can generally be categorized into three major paradigms: model-based methods, including classical machine learning [15] and deep neural networks [16]–[19], incremental learning-based methods [20], [21], and metric-based methods [22], [23]. Model-based methods based on traditional machine learning typically constrain the representation space of known classes by improving algorithms such as support vector machine (SVM), or enhance model reliability by integrating multiple classifiers. In contrast, methods based on deep neural network-based focus on modifying the output layer, optimizing feature representations, or employing generative models such as generative adversarial networks (GANs) to synthesize unknown class samples for robust recognition. Incremental learning-based methods adapt to novel classes as they are detected by the model, through parameter updates such as thresholds or weights, without requiring complete retraining. And metric-based methods evaluate the likelihood of a test sample belonging to known classes by analyzing its statistical relationships, such as feature similarity [22], reconstruction error [23], or distance metrics [18], [19], with respect to the training data.

Owing to their explicit interpretability and flexibility in handling unseen distributions, metric-based methods are particularly well suited for recognition tasks involving highly diverse and unpredictable low-altitude targets. In particular, distance is one of the most commonly used indicators for measuring similarity in both traditional and deep learning contexts [24]. The surrounding prototype loss (SPL) is proposed in [25] to improve open-set recognition performance. It computes the Euclidean distance between sample features and known-class prototype centers, and introduces a surrounding constraint that pushes the prototypes away from the global center. This forces known-class features toward the boundary of the feature space, reducing their overlap with unknown-class features that tend to cluster near the center. Based on K-Contrast Loss, [26] enhances intra-class compactness and inter-class separability by measuring the feature distances between same-class and different-class samples. Combined with deep clustering, it supports both open-set recognition and category discovery for SAR targets. In [27], the multivariate spatial distance is employed to characterize the distribution of radar high-resolution range profile (HRRP) features. Extreme value theory is then applied to construct a decision boundary for detecting unknown targets. The work in [28] models the Euclidean distance between HRRP samples and known-class centers using an extreme value distribution. A tail-based threshold is defined to reject unknown classes, which enables effective open-set recognition.

However, distance-based boundary characterization methods inherently rely on the assumption that class distributions are compact, or can be mapped into compact clusters via deep neural embeddings. In contrast, the radar micro-Doppler feature space naturally exhibits highly nonlinear and non-convex geometries. Consequently, distance-based approaches fail to accurately identify true boundary samples, which ultimately leads to a significant degradation in open-set recognition performance.

To address the limitations mentioned above, this article proposes an open-set recognition framework for rotorcraft UAV classification based on radar micro-Doppler signatures. First, the identification of boundary points for each known class is incorporated, which facilitates open-set recognition decisions. To better capture complex and nonlinear class boundary structures, a novel boundary point identification method based on the minimum description length (MDL) principle is introduced. Furthermore, through theoretical derivation, we prove that the identified boundary points follow the generalized extreme value (GEV) distribution, which in turn enables the construction of statistically sound decision thresholds. Experiments on real-world radar data confirm that the proposed framework effectively models UAV class boundaries and accurately fits the GEV distribution. As a result, it achieves superior performance compared with several state-of-the-art methods in distinguishing rotorcraft UAVs from other unknown aerial targets.

The rest of the article is organized as follows. Section II describes signal model of low-altitude targets and feature extraction. Section III details the boundary point identification based on code length and GEV fitting. Section IV presents the experiments and results based on real-world radar datasets. Finally, Section V concludes this article.

II. Signal Model and Feature Extraction

This section outlines the extraction of micro-Doppler features from radar echoes of rotorcraft UAVs or other aerial targets. The process involves modeling the radar echo with micro-motion components, computing its time-frequency representation, and then extracting dis-

criminative micro-Doppler features.

As shown in Figure 1, the radar is positioned to monitor low-altitude airspace where rotorcraft UAVs, birds, and other aerial targets are likely to appear. Consider a frequency-modulated continuous wave (FMCW) radar system whose transmitted signal can be expressed as follows:

$$s_{\mathrm{tx}}(t) = \exp\left[\mathrm{j}2\pi\left(f_0 t + \frac{K}{2}t^2\right)\right] \tag{1}$$

where f_0 is the starting frequency, $K = B/T$ is the modulation slope, B is the bandwidth, and T is the chirp duration.

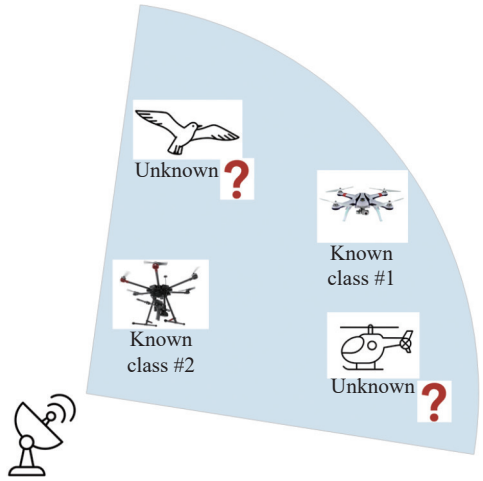


Figure 1 Illustration of the radar observation scenario.

Assuming a micro-motion target such as a flapping bird or rotorcraft UAV with compensated translational motion, the instantaneous range to the target is modeled as

$$R(t) = R_0 + \Delta R(t) \tag{2}$$

where R_0 is the nominal static range and $\Delta R(t)$ represents small oscillations induced by micro-motion. The corresponding time delay is

$$\tau(t) = \frac{2R(t)}{c} = \tau_0 + \delta_\tau(t) \tag{3}$$

where $\tau_0 = 2R_0/c$ and $\delta_\tau(t) = 2\Delta R(t)/c$.

After mixing, the received intermediate frequency (IF) signal becomes

$$s_{\mathrm{IF}}(t) = \alpha \cdot \exp\left\{\mathrm{j}2\pi\left[K\tau(t)t + f_0\tau(t) - \frac{K}{2}\tau^2(t)\right]\right\} \tag{4}$$

where α is the reflection coefficient, and the third term $\frac{K}{2}\tau^2(t)$ is a second-order small quantity and can be neglected. Then the phase can be simplified as

$$\phi(t) \approx K\tau(t)t + f_0\tau(t) \tag{5}$$

Substituting (3) into (5), its two terms become

$$K\tau(t)t = \frac{2Kt}{c}(R_0 + \Delta R(t)) \approx \frac{2Kt}{c}R_0 \tag{6}$$

$$f_0\tau(t) = \frac{2f_0}{c}(R_0 + \Delta R(t)) = \frac{2f_0 R_0}{c} + \frac{2f_0}{c}\Delta R(t) \tag{7}$$

Clearly, equation (6) reflects the target range R_0, whereas equation (7) contains the micro-motion component $\Delta R(t)$.

To localize the target, a fast Fourier transform (FFT) is applied on each chirp frame $s_n(t)$ as follows:

$$S_n(f_r) = \mathrm{FFT}\{s_n(t)\} \tag{8}$$

where f_r corresponds to range bins. The target range bin $[r_1, r_2]$ is selected by detecting significant energy peaks.

To enhance the signal-to-noise ratio of the micro-motion component, the signals from the detected range bins are summed over range to form a micro-motion time series:

$$x(t) = \sum_{r=r_1}^{r_2} S_n(f_r) \tag{9}$$

Then, a short-time Fourier transform (STFT) is applied to analyze the time-frequency structure of the micro-motion signal as follows:

$$X(t, f) = \int_{-\infty}^{\infty} x(\tau)\, w(\tau - t)\, \mathrm{e}^{-\mathrm{j}2\pi f\tau}\, \mathrm{d}\tau \tag{10}$$

where $w(\cdot)$ is a sliding window function. The resulting spectrogram will reveal temporal variations in frequency caused by micro-Doppler effects. For UAVs, rotor-induced signatures appear as harmonics or comb structures, while birds exhibit lower-frequency, more non-stationary modulations due to irregular wing flapping.

Discriminative features are extracted from time-frequency representation to characterize the micro-motion signatures of targets. As the separability of features significantly influences recognition performance [29], a comprehensive set of candidate features is initially extracted to ensure descriptive completeness. Subsequently, feature selection techniques, such as filter methods, are employed to identify the most informative and class-discriminative subsets. A summary of the extracted features is provided in Table 1 [30]–[32].

III. Proposed Method

This section introduces an open-set recognition method for rotorcraft UAV identification, where the overall processing framework is illustrated in Figure 2. After extracting the micro-Doppler features of known UAV classes, identifying boundary points becomes crucial for achieving open-set recognition. To this end, we propose a boundary point identification method based on the MDL principle, which provides a more accurate characterization of class boundaries. Then, the identi-

Table 1 Extracted micro-Doppler features from time-frequency spectrogram

No.	Feature name	Description
1	Mean amplitude	Global mean energy
2	Standard deviation	Global energy spread
3	Skewness	Asymmetry of amplitude distribution
4	Kurtosis	Peakedness of amplitude distribution
5	Entropy	Spectral complexity
6	Singular value decomposition (SVD) energy ratio	Ratio of the first singular value
7	SVD entropy	Entropy of singular value spectrum
8	Row-wise mean energy	Mean temporal energy profile
9	Row-wise energy standard deviation	Standard deviation of temporal energy profile
10	Column-wise mean energy	Mean frequency energy profile
11	Column-wise energy standard deviation	Standard deviation of frequency energy profile
12	Max row index	Peak location (frequency axis)
13	Max column index	Peak location (time axis)
14	Median amplitude	Robust central tendency
15	Interquartile range (IQR)	Robust measure of spread

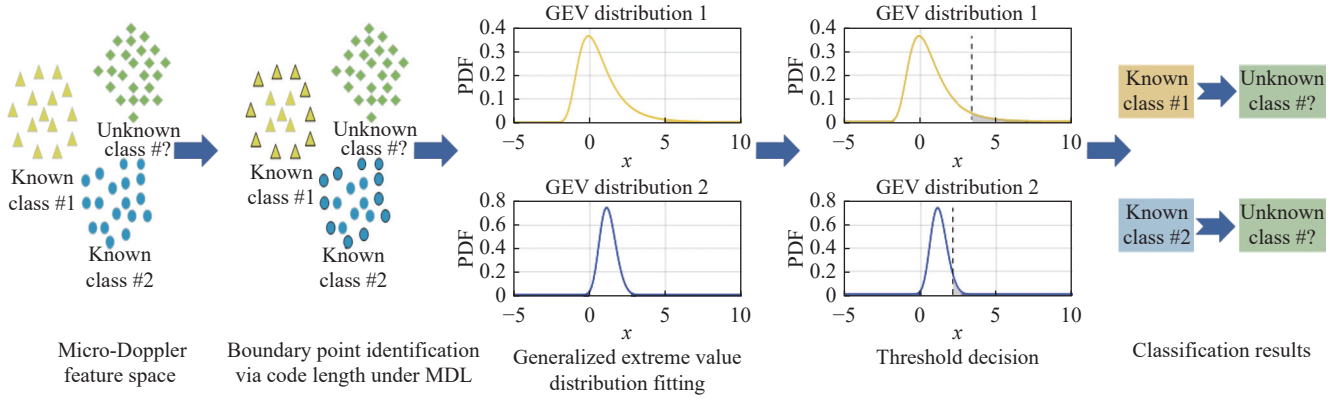


Figure 2 Framework of the proposed open-set UAV recognition method. PDF: probability density function.

fied boundary points are shown to follow the GEV distribution, and this statistical modeling facilitates the establishment of effective decision thresholds. Specifically, a sample is first assigned to the most likely known class, and subsequently a threshold-based decision determines whether it should be rejected as an unknown class.

1. Boundary point identification via code length under MDL

Assume that the micro-Doppler feature training data consist of C known UAV classes. The training set can then be partitioned into C disjoint subsets according to UAV class labels:

$$\boldsymbol{X} = \boldsymbol{X}_1 \cup \boldsymbol{X}_2 \cup \cdots \cup \boldsymbol{X}_C \tag{11}$$

where $\boldsymbol{X}_c = \{\boldsymbol{x}_i \mid y_i = c\}$ represents the subset corresponding to class $c \in \{1, 2, \ldots, C\}$, and y_i denotes the label and $\boldsymbol{X}_i \cap \boldsymbol{X}_k = \varnothing$ for $i \neq k$. Each class c is assumed to follow an underlying class-conditional distribution $P_c(\boldsymbol{x})$. Based on $P_c(\boldsymbol{x})$, the probability density of each training sample within class c can be evaluated. Samples with relatively low density values are likely to lie near the boundary of the class distribution [33].

Given a new sample $\boldsymbol{x}_0$, its likelihood under the distribution $P_c(\boldsymbol{x}_0)$ can be estimated. The resulting probability density indicates the degree to which $\boldsymbol{x}_0$ conforms to the typical characteristics of class c. Specifically, a lower value of $P_c(\boldsymbol{x}_0)$ implies that the sample is located near the boundary or outside the support of the known distribution. This class-conditional density provides a principled way to assess individual samples, especially under open-set recognition scenarios. For bird targets or other low-altitude objects, the micro-Doppler characteristics often differ from those of UAVs, which is likely to result in very low likelihoods under the UAV class-conditional distribution.

While the class-conditional probability density formulation provides an intuitive measure of how likely a sample is to belong to a particular class, it requires knowledge of the true distribution $P_c(\boldsymbol{x})$, which is often intractable in practice. In contrast, the principle of MDL offers a practical and theoretically grounded alternative by approximating the likelihood through the code length induced by a learned probabilistic model. This allows the likelihood of a sample belonging to a

given class to be quantified in terms of the number of bits required to encode it under the corresponding class-conditional model, making the method particularly suitable for implementation in recognition systems.

According to MDL principle, the optimal model M_c for describing class dataset $\boldsymbol{X}_c$ is the one that minimizes the total length of the code required to encode both the model and the data. The fundamental MDL formulation can be expressed as [34]

$$L(\boldsymbol{X}_c, M_c) = L(M_c) + L(\boldsymbol{X}_c \mid M_c) \tag{12}$$

where $L(M_c)$ denotes the code length used to describe the model structure or parameters, and $L(\boldsymbol{X}_c \mid M_c)$ represents the code length of the data under model M_c, which corresponds to the negative log-likelihood if M_c is probabilistic.

In many applications, the model M_c is assumed to be fixed or pretrained, allowing us to treat $L(M_c)$ as a constant. Under this assumption, minimizing the total description length becomes equivalent to minimizing the conditional code length $L(\boldsymbol{X}_c \mid M_c)$.

Evidently, this formulation naturally enables the evaluation of the coding length of an individual sample $\boldsymbol{x}$ under a given class-conditional model. Define the MDL-based code length of $\boldsymbol{x}$ under a given class c:

$$L(\boldsymbol{x} \mid M_c) := -\log_2 P(\boldsymbol{x} \mid M_c) \tag{13}$$

Therefore, the MDL principle provides a tractable formulation for estimating likelihood. As for the underlying data model M, it can be effectively modeled using expressive generative frameworks such as normalizing flows.

After computing the code length for all samples in $\boldsymbol{X}_c$, those with the highest values of code length can be regarded as boundary points of class c. Compared with traditional distance-based heuristics, which often fail under non-convex or complex distributions, the MDL-based criterion is grounded in probability statistics and adapts to the learned data geometry. Therefore, it provides a principled and robust mechanism for detecting boundary points, particularly in the micro motion feature space of UAVs where class boundaries are intricate and uncertain.

2. GEV distribution fitting

To establish a robust rejection mechanism for unknown classes, it is essential to model the statistical behavior of boundary points. As previously defined, $L(\boldsymbol{x} \mid M_c)$ measures the code length of a sample under the model of class c. Typically, samples residing near the class center exhibit small values of $L(\boldsymbol{x} \mid M_c)$, whereas those situated closer to the boundary or originating from outside the class tend to produce substantially larger values [34]. Empirical studies and theoretical considerations indicate that the distribution of $L(\boldsymbol{x} \mid M_c)$ often exhibits heavy-tailed or right-skewed characteristics, with boundary values correspond to structurally sensitive regions of the class distribution, which are characterized by low density and play a critical role in determining the stability of decision boundaries and the generalization performance of the model [35].

To rigorously characterize the statistical behavior of high-code-length samples, this work adopts the framework of extreme value theory (EVT) [36], under which the distribution of maximum code length is shown to follow the GEV family [37].

Consider a known class c with training sample set $\boldsymbol{S} = \{\boldsymbol{s}_1, \boldsymbol{s}_2, \ldots, \boldsymbol{s}_n\}$. For each candidate sample $\boldsymbol{s}_i$, the code length with respect to the class model is computed as

$$L(\boldsymbol{s}_i \mid M_c) = -\log_2 P(\boldsymbol{s}_i \mid M_c) \tag{14}$$

resulting in a sequence of code length values $\{L_1, L_2, \ldots, L_n\}$. Let the top-$k$ code lengths be denoted as

$$\boldsymbol{A}_k = \left\{L_{(1)}, L_{(2)}, \ldots, L_{(k)}\right\} \tag{15}$$

where $L_{(1)} \geq L_{(2)} \geq \cdots \geq L_{(k)}$ are the k largest values in the code length sequence. These samples are correspond to structurally sensitive points near class boundaries.

First, the training samples $\boldsymbol{s}_i \in \boldsymbol{S}$ are assumed to be independently and identically distributed (i.i.d.) realizations drawn from the true distribution of class c. Due to the definition of the code length and M_c denotes a fixed class-conditional model trained on samples from c, the resulting sequence of code lengths inherits the i.i.d. property. That is, since each L_i is obtained by applying a fixed mapping to an i.i.d. sample $\boldsymbol{s}_i$, the sequence $\{L_i\}_{i=1}^n$ also forms an i.i.d. set of random variables.

According to the Fisher-Tippett-Gnedenko theorem [38], the maximum of a sequence of i.i.d. random variables—when properly normalized—converges in distribution to a GEV distribution. Specifically, let $\{X_i\}$ be i.i.d. variables and let $M_k = \max\{X_1, X_2, \ldots, X_k\}$; then there exist sequences of normalizing constants $\{a_k > 0\}$ and $\{b_k \in \mathbb{R}\}$ such that:

$$\frac{M_k - b_k}{a_k} \xrightarrow[k\to\infty]{\mathscr{D}} G(x; \mu, \sigma, \xi) \tag{16}$$

where $G(x)$ denotes a GEV distribution parameterized by location μ, scale σ, and shape ξ. The family encompasses three limiting forms —Gumbel, Fréchet, and Weibull—unified by the following expression:

$$G(x; \mu, \sigma, \xi) = \begin{cases} \exp\left(-\exp\left(-z\right)\right), & \xi = 0 \\ \exp\left(-(1-\xi z)^{1/\xi}\right), & \xi \neq 0,\ 1-\xi z > 0 \end{cases} \tag{17}$$

where $z = (x-\mu)/\sigma$, and the distribution type is determined by the shape parameter ξ as follows: Gumbel when $\xi = 0$, Fréchet when $\xi > 0$, and Weibull when $\xi < 0$.

Therefore, the top-k code lengths $\boldsymbol{A}_k$, can be regarded as order statistics from the tail of an i.i.d. sequence. As $n \to \infty$, the distribution of the maximum $\Delta L_{(1)}$ asymptotically follows a GEV law, and the joint distribution of the top-k values is also governed by extreme value behavior.

Moreover, each code length is inherently non-negative, since the conditional probability $P(\boldsymbol{s}_i \mid M_c)$ lies strictly within the interval $(0, 1]$. As a result, the support of the code length distribution is bounded below by zero. This boundedness places structural constraints on the admissible form of the limiting GEV distribution. Specifically, depending on the tail behavior of the distribution, the appropriate subclass must be either the Fréchet distribution (corresponding to a heavy-tailed case with $\xi > 0$) or the Weibull distribution (corresponding to a bounded-tailed case with $\xi < 0$).

Therefore, the boundary value characteristics of code length satisfies the theoretical conditions of GEV convergence. Once a GEV distribution is fitted to the top-k code lengths for each known class during training, the resulting distribution can be used to establish a class-specific rejection threshold.

Let $\boldsymbol{x}_{\text{test}}$ be a test sample. For each known class $c \in \{1, 2, \ldots, C\}$, its code length is computed as

$$L(\boldsymbol{x}_{\text{test}} \mid M_c) = -\log_2 P(\boldsymbol{x}_{\text{test}} \mid M_c) \tag{18}$$

The predicted class is selected as

$$c^* = \arg\min_c L(\boldsymbol{x}_{\text{test}} \mid M_c) \tag{19}$$

To determine whether the sample belongs to the known class c^* or should be rejected as unknown, the corresponding code length is compared against the α-quantile threshold of the fitted GEV distribution:

$$\tau_{c^*} = G^{-1}(\alpha; \mu_{c^*}, \sigma_{c^*}, \xi_{c^*}) \tag{20}$$

where G^{-1} denotes the inverse cumulative distribution function (CDF) of the GEV distribution fitted to class c^*. If

$$L(\boldsymbol{x}_{\text{test}} \mid M_{c^*}) > \tau_{c^*} \tag{21}$$

then the test sample is rejected as an outlier or unknown instance; otherwise, it is accepted as belonging to class c^*.

Finally, the complete open set recognition procedure based on MDL-driven code length and GEV modeling is summarized in Algorithm 1.

Algorithm 1 Open set recognition via MDL-based code length

Input: Training set $\boldsymbol{X} = \{\boldsymbol{x}_1, \boldsymbol{x}_2 \ldots, \boldsymbol{x}_N\}$, class labels $\boldsymbol{y}$, generative model, top-k parameter, GEV confidence level α.

Output: Class-specific thresholds τ_c, open set classifier.

Preprocessing: Apply feature selection on training set $\boldsymbol{X}$ to obtain $\boldsymbol{X}'$;
Training phase: foreach class $c \in \{1, 2, \ldots, C\}$ **do**
  Extract samples $\boldsymbol{S}_c = \{\boldsymbol{s}_i \in \boldsymbol{X}' \mid y_i = c\}$;
  Train probabilistic model M_c on $\boldsymbol{S}_c$;
  Compute code lengths:
    $L_i = -\log_2 P(\boldsymbol{s}_i \mid M_c)$ for all $\boldsymbol{s}_i \in \boldsymbol{S}_c$;
  Sort L_i in descending order to get top-k values $\boldsymbol{A}_k^{(c)}$;
  Fit GEV distribution $G_c(x; \mu_c, \sigma_c, \xi_c)$ to $\boldsymbol{A}_k^{(c)}$;
  Compute rejection threshold:
  $\tau_c = G_c^{-1}(\alpha; \mu_c, \sigma_c, \xi_c)$;
end
Inference phase: foreach test sample $\boldsymbol{x}_{\text{test}}$ **do**
  Apply feature selection to obtain $\boldsymbol{x}'_{\text{test}}$;
  Compute code lengths:
    $L = -\log_2 P(\boldsymbol{x}'_{\text{test}} \mid M_c)$ for each known class c;
  Assign class label:
    $c^* = \arg\min_c L_c$;
  if $L(\boldsymbol{x}'_{\text{test}} \mid M_{c^*}) > \tau_{c^*}$ **then**
    Reject as `unknown`;
  else
    Accept as class c^*;
  end
end

IV. Experiments

This section experimentally validates the effectiveness of the proposed approach. We first introduce the datasets used in our evaluation. Subsequently, boundary point identification and GEV fitting experiments are conducted to demonstrate the effectiveness of our MDL-based modeling. Finally, comparisons with state-of-the-art open-set recognition methods highlight the superiority of ours.

1. Dataset introduction

This study employs the publicly available multi-band and multi-angle FMCW radar dataset (LSS-FMCWR-2.0) [39], focusing specifically on the K-band subset. Four representative low-altitude and slow-speed aerial targets are selected, including DJI Inspire 2 (DJI2), a hexarotor UAV (HexUAV), the AC311 helicopter, and a bionic flapping-wing bird. As illustrated in Figure 3, these targets encompass typical multi-rotor, single-rotor, and flapping-wing aerial platforms. The key physical parameters of the four targets are summarized in Table 2.

The radar operates at 23.7 GHz and captures fine-grained micro-motion signatures such as rotor harmonics and flapping-induced frequency modulation. Table 3 summarizes the core radar configuration when obtaining radar data.

For the multirotor UAV targets (DJI2 and HexUAV), each observation angle includes three independent measurement trials. Among them, the first two trials are used for training, and the third is reserved for testing. In contrast, the AC311 helicopter and the bion-

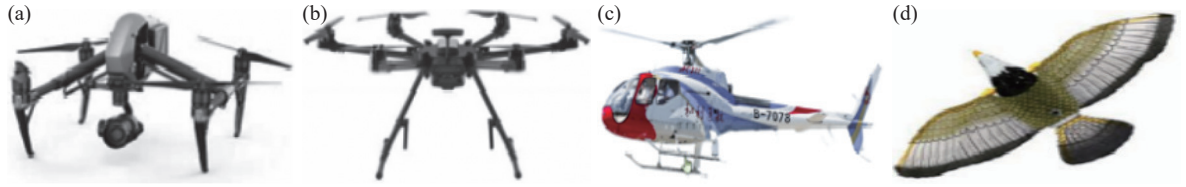


Figure 3 Visual examples of the four targets: (a) DJI Inspire 2; (b) Hexarotor UAV; (c) AC311 helicopter; (d) Bionic flapping-wing bird.

Table 2 Specification parameters of four targets (Refer to [39] for the dataset LSS-FMCWR-2.0 on the K-band subset)

Four targets in this paper	Size (mm)	RCS (dBsm)
DJI2	605 × 588 × 315	−20 to −13
HexUAV	800 to 1200	−5 to 3
AC311	–	−15 to −8
Bionic bird	500 to 1500	<−30

ic flapping-wing bird have two trials, both of which are used exclusively for testing.

Due to the long duration of radar measurement, a sliding window approach is applied to segment the time-frequency spectrograms. The representative time-frequency spectrograms of the four target classes are illustrated in Figure 4, highlighting characteristic micro-Doppler patterns such as rotor harmonics and non-periodic flapping motion. Based on the spectrograms, a se-

Table 3 Radar parameter settings for four targets

Target	Frequency (GHz)	Bandwidth (MHz)	Sampling rate (kHz)	Chirp period (ms)	Distance (m)	Angles (°)
DJI2	23.7	100	500	0.3	8	0, 60, 90, 120, 180
HexUAV	23.7	100	500	0.3	8	0, 60, 90, 120, 180
AC311	23.7	100	500	4	32	–
Bionic bird	23.7	200	500	0.3	2	–

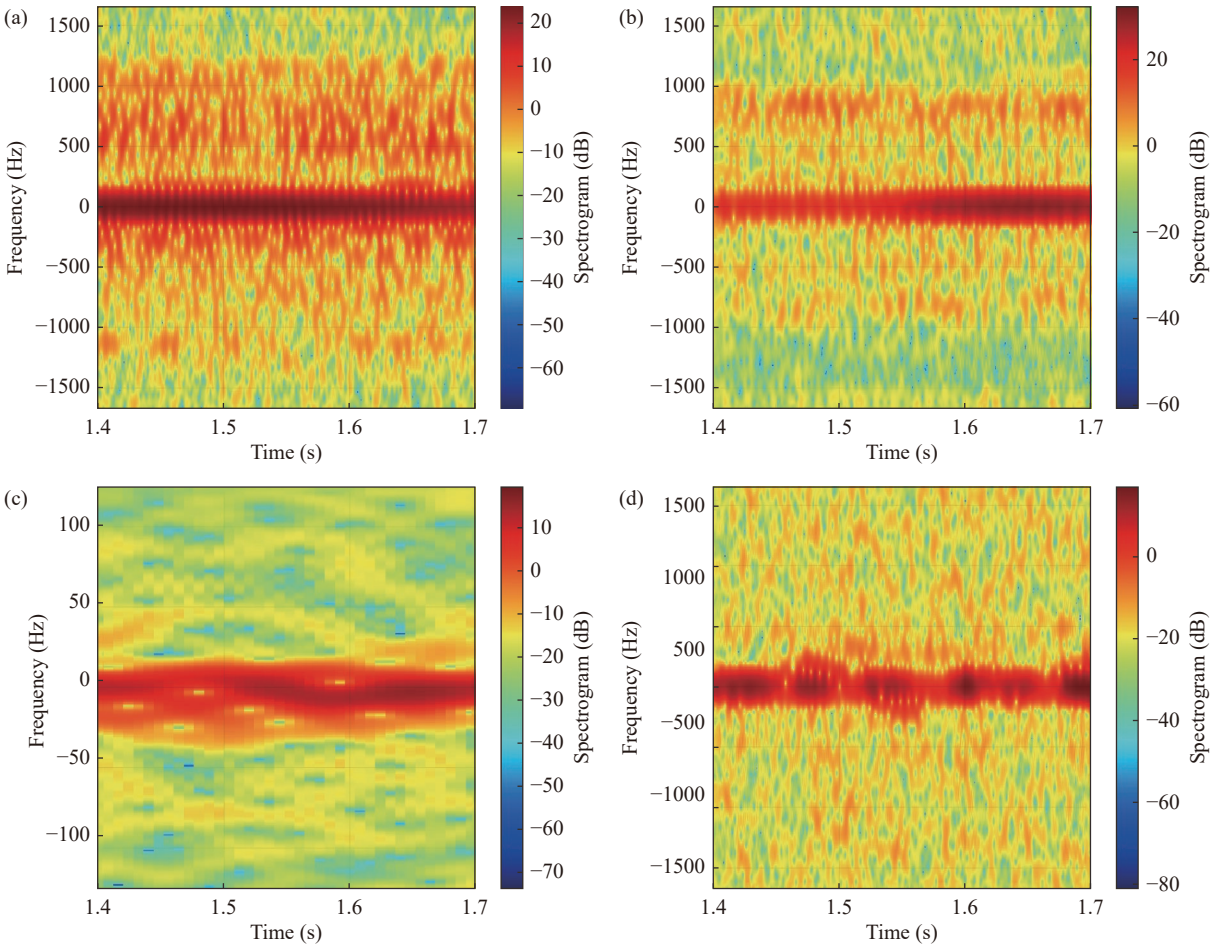


Figure 4 Time-frequency spectrograms of the four targets: (a) DJI Inspire 2; (b) Hexarotor UAV; (c) AC311 helicopter; (d) Bionic flapping-wing bird.

ries of handcrafted time-frequency domain features, as summarized in Table 1, are extracted from each windowed segment. Subsequently, the dataset partitioning is summarized in Table 4.

Table 4 Dataset split of four targets

Class	Type	Training	Testing
Known	DJI2	1332	666
	HexUAV	1110	555
Unknown	AC311	0	216
	Bionic bird	0	222

The DJI2 and HexUAV were treated as two different known classes, and AC311 and bionic bird were regarded as unknown classes. In the experiments, the training set contained only the known classes, while the test set included both known and unknown classes. After feature extraction, a filter-based feature selection method using univariate statistical tests was employed to select the optimal feature subset for subsequent experiments. To determine the optimal number of features, a 5-fold stratified cross-validation was conducted on the training set to calculate the mean validation accuracy (CV_Acc) under different numbers of selected features. As illustrated in Figure 5, the average cross-validation accuracy increases steadily as the number of features increases from 4 to 10. The principle for feature number selection is to achieve the highest possible recognition accuracy by utilizing the smallest number of features. Therefore, 10 features are selected as the optimal feature dimension, which provides a favorable trade-off between recognition performance and feature compactness.

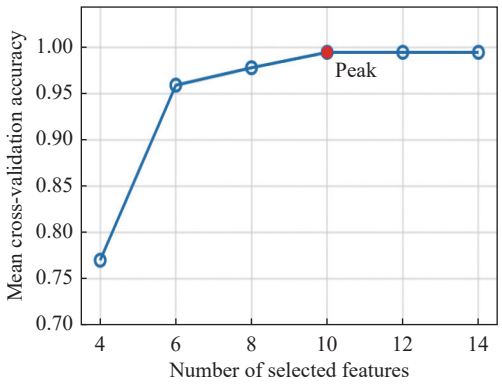


Figure 5 Cross-validation accuracy under different numbers of selected features.

2. Boundary point identification

This section aims to validate the effectiveness of the proposed MDL-based boundary point identification method.

During training, only the known class samples are used to fit a generative model based on RealNVP [40], an invertible flow architecture. Following training, the code length of each sample is computed according to the MDL principle. The top 10% of samples with the highest code length are identified as MDL-based boundary points. For comparison, three distance-based boundary sets are extracted by selecting the top 10% of training samples with the largest Euclidean, Mahalanobis and Cosine distance from the class center.

Figure 6 shows the t-distributed stochastic neighbor embedding (t-SNE) [41] visualization of boundary points extracted by the distance-based and the proposed MDL-based methods. As shown in Figures 6(a)–6(c), the distance-based methods tend to misidentify internal samples as boundary points, since it relies purely on geometric proximity without considering the underlying data distribution. This often results in scattered boundary assignments within the class clusters. In contrast, the MDL-based method in Figure 6(d) yields boundary points that are more coherently aligned with the actual class margins, effectively capturing complex and nonlinear class structures. Such precise boundary identification is crucial for open-set recognition, as it enhances the ability to distinguish known classes from potential unknown targets.

Furthermore, Figure 7 presents the response distributions of unknown-class test samples against the two known-class models, in terms of both code length and distance metrics. The code length values of unknown samples are significantly higher than those of known classes, indicating greater encoding complexity and better separation capability. In contrast, the three distance distributions show considerable overlap between known and unknown classes, limiting their open-set discrimination effectiveness.

These results demonstrate that the MDL-based method provides more accurate boundary characterization in open-set radar UAV target recognition scenarios.

3. GEV fitting

After the boundary points are extracted based on the MDL-driven code length criterion, the GEV distributions are subsequently fitted to their corresponding values. To validate this modeling choice, we further compare GEV against representative alternatives: Gaussian as a light-tailed distribution, Log-normal as a medium-tailed distribution, and Student-t as a heavy-tailed distribution [42].

Figure 8 illustrates the fitted probability density functions (PDFs) of different distributions superimposed on the boundary histograms of the DJI Inspire 2 and Hexrotor UAVs. As shown in both subfigures, the Gaussian distribution underestimates the probability mass in the tail regions, reflecting its limitation as a light-tailed model. The Log-normal and Student-t distributions capture heavier tails than Gaussian, yet still fail to align with the sharp decay patterns of the boundary points. In contrast, the GEV distribution provides a much closer fit across both the bulk and the extreme regions, especially near the right tails where boundary points concentrate.

In addition, to quantitatively assess the goodness-

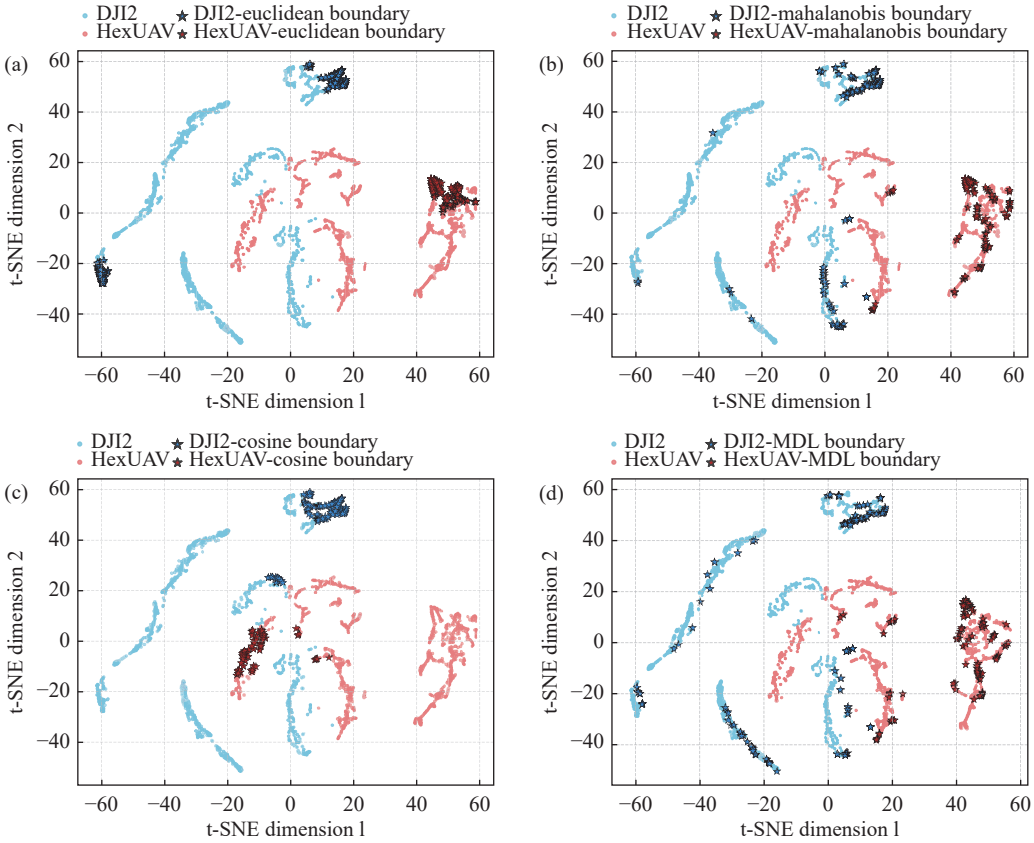


Figure 6 t-SNE visualization of boundary points identified by distance-based and MDL-based methods: (a) Euclidean distance-based; (b) Mahalanobis distance-based; (c) Cosine distance-based; (d) MDL-based.

of-fit, we performed Kolmogorov-Smirnov (KS) tests under different distributional assumptions. The results, summarized in Table 5, show that for both UAV classes, the GEV distribution yields the smallest KS statistics along with p-values well above the 0.05 significance level, supporting the hypothesis that boundary points follow a GEV principle. In contrast, Gaussian, Log-normal, and Student-t distributions exhibit significantly larger KS values and p-values below 0.05, indicating a poor match to the boundary.

These observations validate the theoretical assumption that boundary points exhibit extreme-value characteristics, thereby supporting the adoption of GEV as a statistically sound modeling choice.

Additionally, the rejection thresholds were statistically determined from the fitted GEV distributions of MDL-based code lengths for each known class at a 95% confidence level. Specifically, the thresholds were 9.45 for DJI Inspire 2 and 9.59 for Hexarotor UAV, corresponding to the 95th percentile of their respective boundary code-length distributions.

4. Recognition performance

In this section, we evaluate the OSR performance of the proposed method. Several classical and representative baselines are adopted for comparison, including OCSVM [43], OpenMax [16], ARPL [44], OSRCI [45], CGDL [46], MAHA [47], and distance-based thresholding (DIS) [28].

To comprehensively assess recognition performance under open-set scenarios, a set of evaluation metrics is employed. In this context, TP (true positive) and FN (false negative) represent known samples that are correctly and incorrectly classified, respectively, whereas TN (true negative) and FP (false positive) represent unknown samples that are correctly rejected and incorrectly accepted.

- Accuracy (Acc) [48]: Overall classification accuracy:

$$\mathrm{Acc} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN}}$$

- Closed-set recognition accuracy (CSR_Acc) [48]: Accuracy on known classes:

$$\mathrm{CSR}_{\mathrm{Acc}} = \frac{\mathrm{TP}_{\mathrm{known}}}{\mathrm{TP}_{\mathrm{known}} + \mathrm{FN}_{\mathrm{known}}}$$

- Unknown-class accuracy (UK_Acc) [48]: Correct rejection rate of unknown samples:

$$\mathrm{UK}_{\mathrm{Acc}} = \frac{\mathrm{TN}_{\mathrm{unknown}}}{\mathrm{TN}_{\mathrm{unknown}} + \mathrm{FP}_{\mathrm{unknown}}}$$

- Macro average F1 score (MAF1) [49]: Average of F1 scores for known and unknown classes:

$$\mathrm{MAF1} = \frac{1}{2}\left(\mathrm{F1}_{\mathrm{known}} + \mathrm{F1}_{\mathrm{unknown}}\right)$$

with

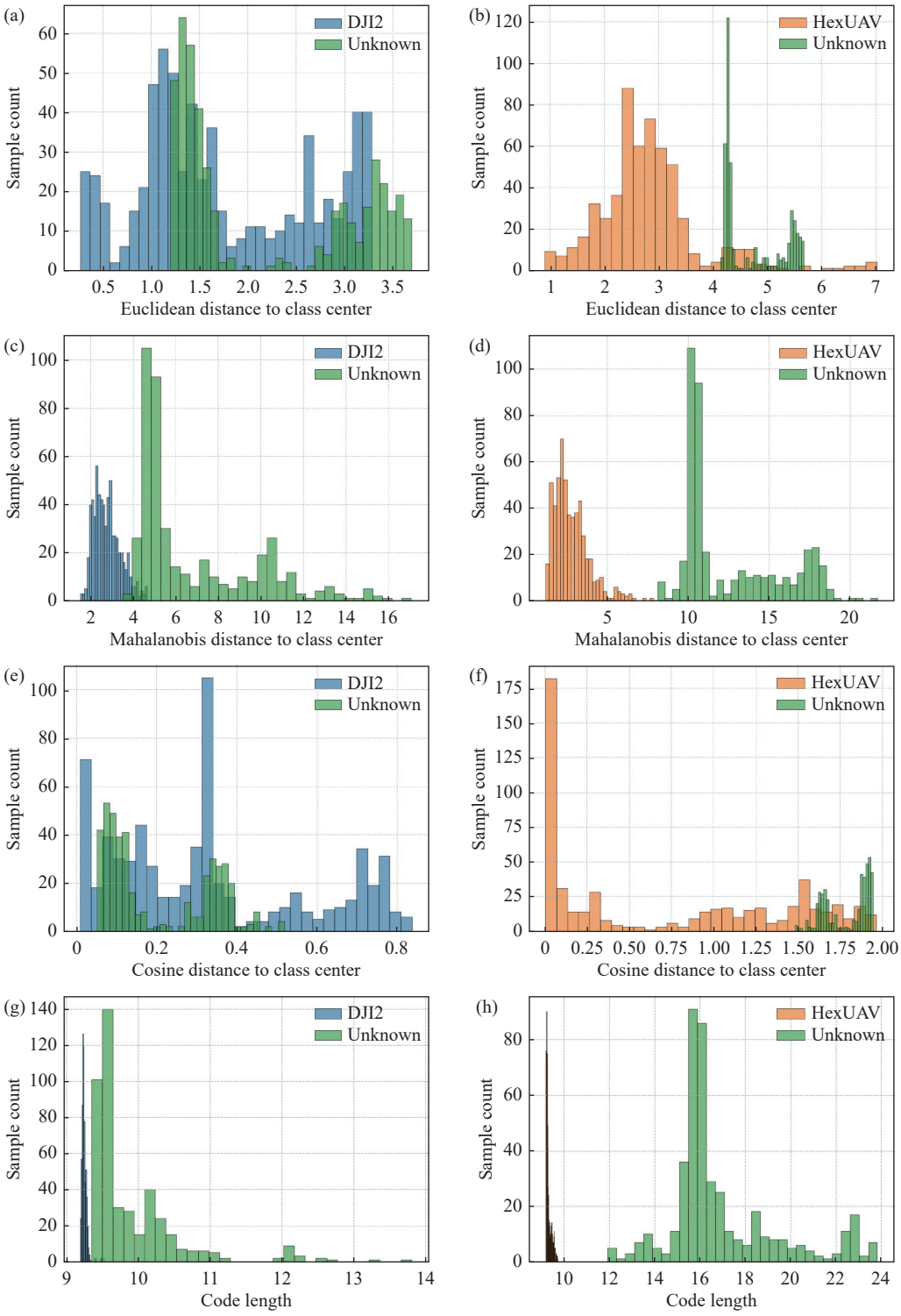


Figure 7 Distributions of model responses for test samples using distance-based and MDL-based methods: (a) Euclidean distance-based responses of the DJI2 model to DJI2 and unknown-class samples; (b) Euclidean distance-based responses of the HexUAV model to HexUAV and unknown-class samples; (c) Mahalanobis distance-based responses of the DJI2 model to DJI2 and unknown-class samples; (d) Mahalanobis distance-based responses of the HexUAV model to HexUAV and unknown-class samples; (e) Cosine distance-based responses of the DJI2 model to DJI2 and unknown-class samples; (f) Cosine distance-based responses of the HexUAV model to HexUAV and unknown-class samples; (g) MDL-based responses of the DJI2 model to DJI2 and unknown-class samples; (h) MDL-based responses of the HexUAV model to HexUAV and unknown-class samples.

$$\mathrm{F1}_c = \frac{2 \cdot P_c \cdot R_c}{P_c + R_c},\ P_c = \frac{\mathrm{TP}_c}{\mathrm{TP}_c + \mathrm{FP}_c},\ R_c = \frac{\mathrm{TP}_c}{\mathrm{TP}_c + \mathrm{FN}_c}$$

where $c \in \{\text{known}, \text{unknown}\}$.

- AUROC [50]: Area under the receiver operating characteristic (ROC) curve for distinguishing between known and unknown classes.
- FPR@95TPR [50]: False positive rate when the true positive rate reaches 95%:

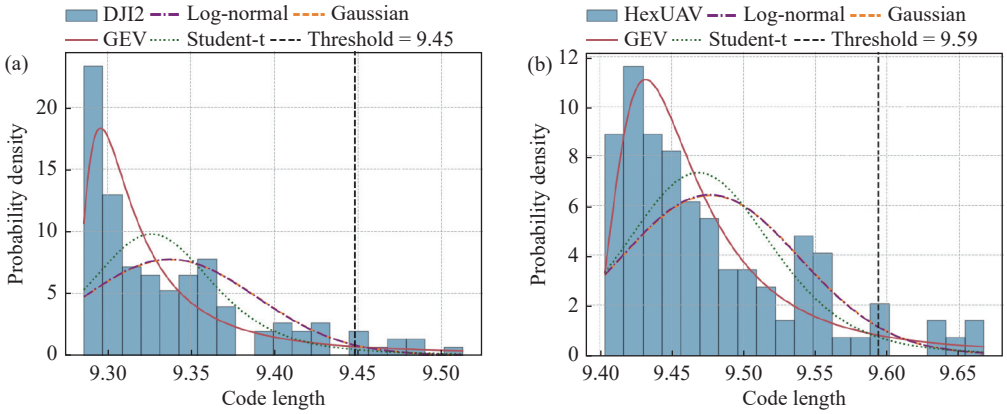


Figure 8 Fitting results of different distributions to boundary points using MDL-based methods: (a) Fitting result of DJI2; (b) Fitting result of HexUAV.

Table 5 KS test results of boundary points fitting under different distributions

Target	Different distribution assumption	KS	p-value
DJI2	Gaussian	0.1392	0.0100
	Log-normal	0.1386	0.0105
	Student-t	0.1351	0.0135
	GEV	**0.0932**	**0.1826**
HexUAV	Gaussian	0.2033	0.0002
	Log-normal	0.2013	0.0002
	Student-t	0.2023	0.0002
	GEV	**0.0655**	**0.7028**

$$\mathrm{FPR@95TPR} = \frac{\mathrm{FP}}{\mathrm{FP}+\mathrm{TN}}\Big|_{\mathrm{TPR}\geq 95\%}$$

Table 6 presents a comprehensive comparison across all methods and metrics. Overall, the proposed method achieves the best performance compared with other baselines, where the CSR_Acc is comparable to the best baseline, while the other metrics consistently outperform all alternatives. It demonstrates strong recognition capability for known classes while effectively rejecting unknowns, achieving a well-balanced trade-off between closed-set and open-set performance. In contrast, several baseline methods struggle with unknown-class detection, leading to degraded overall performance.

Table 6 Open-set recognition performance of different methods

Method	Acc (%)	CSR_Acc (%)	UK_Acc (%)	MAF1 (%)	AUROC (%)	FPR@95TPR ↓ (%)
OCSVM [43]	71.91	72.15	71.23	62.12	87.56	70.76
OpenMax [16]	70.34	85.75	27.40	47.42	56.07	84.52
ARPL [44]	77.28	89.76	42.47	56.30	83.27	43.73
OSRCI [45]	79.39	**98.77**	25.34	69.60	86.67	60.69
CGDL [46]	87.88	96.81	63.01	70.02	97.74	9.99
MAHA [47]	89.33	90.42	86.30	74.45	98.48	5.81
DIS [28]	57.63	68.47	27.40	41.45	61.01	100.00
Ours	**95.72**	94.35	**99.54**	**79.49**	**99.38**	**1.88**

To verify the effectiveness of the proposed method in multi-class scenarios, two additional UAV types, DJI M350 (M350) and DJI Mavic 2 (DJIM2), were introduced into the experiments. Since this work focuses specifically on UAV recognition, AC311 and Bionic Bird are consistently treated as unknown classes throughout the experiments.

Firstly, Table 7 summarizes the recognition accuracy of different methods under the scenario where the unknown classes are fixed as AC311 and the bionic bird, while the number of known classes increases. It can be observed that the proposed method outperforms the other methods.

Moreover, to demonstrate the effectiveness of the proposed method under different openness levels, the known classes were fixed as DJI2 and HexUAV, while the number of unknown classes gradually increased from one to four. The corresponding openness can be calculated as follows [28]:

$$\mathrm{Openness} = 1 - \sqrt{\frac{2N_{\mathrm{train}}}{N_{\mathrm{test}} + N_{\mathrm{target}}}} \tag{22}$$

where N_{train} denotes the number of training classes, N_{test} represents the total number of classes in the test set (including both known and unknown), and N_{target} indicates the number of target classes to be recognized (usually equal to N_{train}). Table 8 shows that as open-

Table 7 Accuracy results (%) of different methods under varying numbers of known classes

Method	Known class(es)		
	DJI2, HexUAV	DJI2, HexUAV, DJIM2	DJI2, HexUAV, DJIM2, M350
OCSVM	71.91%	60.60%	43.63%
Openmax	70.34%	59.12%	48.60%
ARPL	77.28%	71.56%	68.24%
OSRCI	79.39%	76.34%	73.15%
CGDL	87.88%	81.46%	68.17%
MAHA	89.33%	81.68%	69.17%
DIS	57.63%	35.74%	27.79%
Ours	**95.72%**	**91.70%**	**79.51%**

ness increases, the accuracy of all methods decreases, which is expected since more unknown classes make recognition harder. However, the proposed method consistently outperforms the others and shows the slowest performance degradation, demonstrating better robustness under high-openness conditions.

Lastly, considering that, in practical scenarios, the radar time-frequency representations are inevitably affected by noise and clutter, we further evaluated their impact on the OSR performance. In the experiments, Gaussian noise was considered for the noise component, while the clutter was modeled by a Weibull distribution [51] due to its non-Gaussian and heavy-tailed characteristics arising from complex scattering mechanisms such as ground surfaces and vegetation. The signal-to-noise ratio (SNR) ranged from 10 dB to 20 dB and the signal-to-clutter ratio (SCR) from 20 dB to 30 dB, both varied in 2 dB increments. These disturbances were artificially added to the original time–frequency representations of testing set before feature extraction and classification.

Table 8 Accuracy results (%) of different methods under varying openness

Method	Unknown class(es) under different openness			
	AC311	AC311, Bionic Bird	AC311, Bionic Bird, DJI2	AC311, Bionic Bird, DJI2, M350
	0.106	0.184	0.244	0.293
OCSVM	75.37%	71.91%	62.92%	54.20%
Openmax	74.91%	70.34%	49.98%	38.85%
ARPL	77.34%	77.28%	65.12%	57.64%
OSRCI	83.65%	79.39%	57.08%	47.51%
CGDL	91.52%	87.88%	64.99%	54.23%
MAHA	91.86%	89.33%	77.25%	63.99%
DIS	57.93%	57.63%	41.12%	35.67%
Ours	**95.84%**	**95.72%**	**87.48%**	**76.76%**

Figure 9 presents the overall accuracy of different methods under various SNR and SCR conditions. As the SNR or SCR decreases, a noticeable degradation in performance is observed for all methods. Nevertheless, when the SNR exceeds 10 dB or the SCR exceeds 20 dB, the proposed method consistently achieves higher accuracy than the compared methods, indicating its better robustness to both Gaussian noise and Weibull-modeled clutter.

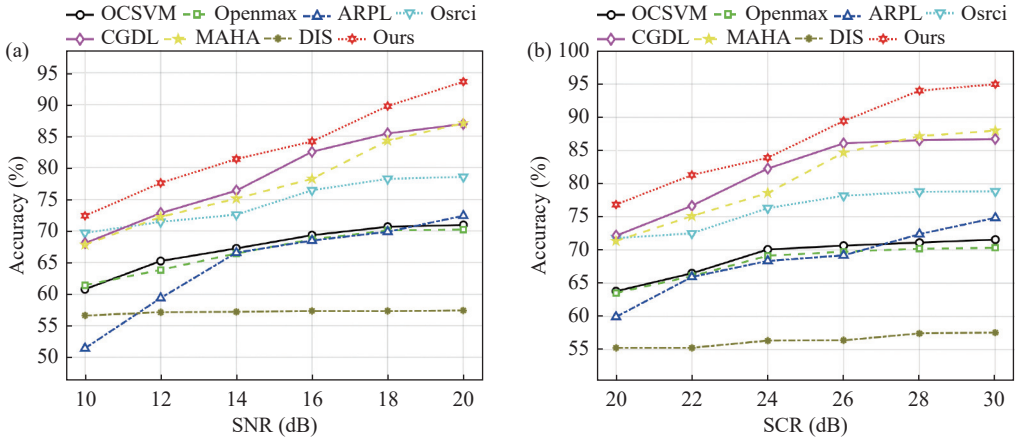


Figure 9 Overall accuracy of different methods under various SNRs and SCRs: (a) Various SNRs; (b) Various SCRs.

V. Conclusion

This paper presents an effective open-set recognition framework for identifying rotorcraft UAVs and other aerial targets in low-altitude UAV surveillance. First, a boundary point identification method based on code length under the MDL principle is proposed to effectively detect complex class boundaries. Moreover, through theoretical analysis, we further prove that these boundary points follow a GEV distribution, thereby supporting the construction of statistically grounded decision thresholds and improving the modeling of unknown classes. Experimental results on real radar datasets validate the effectiveness of the proposed approach, demonstrating significant performance gains over existing open-set recognition methods.

However, our method may struggle when the UAV types exhibit highly similar micro-Doppler signatures or when severe noise/clutter degrades the quality of features. In future work, motivated by the challenges of

real-world low-altitude UAV surveillance —including non-stationary clutter, low and varying SNR/SCR, target maneuvering, and limited availability of large-scale realistic datasets—we will further enhance feature representations, develop more adaptive MDL-based models, and investigate multi-representation fusion, such as combining time-frequency representations with HRRP-based features, to improve robustness under practical operating conditions.

Acknowledgements

This work was supported by the Aeronautical Science Foundation of China (Grant No. 2018ZC51022).

References

[1] B. I. Ahmad, C. Rogers, S. Harman, *et al.*, "A review of automatic classification of drones using radar: Key considerations, performance evaluation, and prospects," *IEEE Aerospace and Electronic Systems Magazine*, vol. 39, no. 2, pp. 18–33, 2024.

[2] M. H. Rahman, M. A. S. Sejan, M. A. Aziz, *et al.*, "A comprehensive survey of unmanned aerial vehicles detection and classification using machine learning approach: Challenges, solutions, and future directions," *Remote Sensing*, vol. 16, no. 5, article no. 879, 2024.

[3] W. Jiang, Z. Liu, Y. P. Wang, *et al.*, "Realizing small UAV targets recognition via multi-dimensional feature fusion of high-resolution radar," *Remote Sensing*, vol. 16, no. 15, article no. 2710, 2024.

[4] B. Torvik, K. E. Olsen, and H. Griffiths, "Classification of birds and UAVs based on radar polarimetry," *IEEE Geoscience and Remote Sensing Letters*, vol. 13, no. 9, pp. 1305–1309, 2016.

[5] M. Ritchie, C. Horne, and N. Peters, "Radar UAV and bird signature comparisons with micro-Doppler," in *Radar Countermeasures for Unmanned Aerial Vehicles*, C. Clemente, F. Fioranelli, F. Colone, *et al.*, Eds. IET, London, UK, pp. 279–314, 2021.

[6] W. S. Chen, J. Liu, and J. Li, "Classification of UAV and bird target in low-altitude airspace with surveillance radar data," *The Aeronautical Journal*, vol. 123, no. 1260, pp. 191–211, 2019.

[7] V. C. Chen, F. Li, S. S. Ho, *et al.*, "Micro-Doppler effect in radar: Phenomenon, model, and simulation study," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 42, no. 1, pp. 2–21, 2006.

[8] D. Tahmoush, "Review of micro-Doppler signatures," *IET Radar, Sonar & Navigation*, vol. 9, no. 9, pp. 1140–1146, 2015.

[9] M. Ritchie, F. Fioranelli, H. Griffiths, *et al.*, "Monostatic and bistatic radar measurements of birds and micro-drone," in *Proceedings of 2016 IEEE Radar Conference (RadarConf)*, Philadelphia, PA, USA, pp. 1–5, 2016.

[10] J. H. Park, Y. H. Jang, S. H. Kang, *et al.*, "Enhanced UAV detection and classification with birds using NLFM pulse-Doppler radar," *IEEE Access*, vol. 13, pp. 93605–93613, 2025.

[11] S. A. Zulkarnain, S. Zulkifli, and 'A. M. Ali, "Identification and analysis of micro-Doppler signature of a bird versus micro-UAV," *Journal of Advanced Research in Micro and Nano Engineering*, vol. 16, no. 1, pp. 102–113, 2024.

[12] X. L. Chen, H. Zhang, J. Song, *et al.*, "Micro-motion classification of flying bird and rotor drones via data augmentation and modified multi-scale CNN," *Remote Sensing*, vol. 14, no. 5, article no. 1107, 2022.

[13] R. M. Narayanan, B. Tsang, and R. Bharadwaj, "Classification and discrimination of birds and small drones using radar micro-Doppler spectrogram images," *Signals*, vol. 4, no. 2, pp. 337–358, 2023.

[14] W. J. Scheirer, A. de Rezende Rocha, A. Sapkota, *et al.*, "Toward open set recognition," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 35, no. 7, pp. 1757–1772, 2013.

[15] H. Cevikalp, "Best fitting hyperplanes for classification," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 39, no. 6, pp. 1076–1088, 2017.

[16] A. Bendale and T. E. Boult, "Towards open set deep networks," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, Las Vegas, NV, USA, pp. 1563–1572, 2016.

[17] R. Yoshihashi, W. Shao, R. Kawakami, *et al.*, "Classification-reconstruction learning for open-set recognition," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, Long Beach, CA, USA, pp. 4011–4020, 2019.

[18] P. R. Mendes Júnior, R. M. De Souza, R. D. O. Werneck, *et al.*, "Nearest neighbors distance ratio open-set classifier," *Machine Learning*, vol. 106, no. 3, pp. 359–386, 2017.

[19] M. Hassen and P. K. Chan, "Learning a neural-network-based representation for open set recognition," in *Proceedings of the 2020 SIAM International Conference on Data Mining*, Cincinnati, OH, USA, pp. 154–162, 2020.

[20] R. De Rosa, T. Mensink, and B. Caputo, "Online open world recognition," *arXiv preprint*, arXiv: 1604.02275, 2016.

[21] V. M. Venkataram, "Open set text classification using neural networks," *Master Thesis*, University of Colorado Colorado Springs, Colorado Springs, CO, USA, 2018.

[22] G. L. Fei and B. Liu, "Social media text classification under negative covariate shift," in *Proceedings of 2015 Conference on Empirical Methods in Natural Language Processing*, Lisbon, Portugal, pp. 2347–2356, 2015.

[23] H. Zhang and V. M. Patel, "Sparse representation-based open set recognition," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 39, no. 8, pp. 1690–1696, 2017.

[24] F. Gao, L. Yang, and H. Li, "A survey on open set recognition," *Journal of Nanjing University (Natural Sciences)*, vol. 58, no. 1, pp. 115–134, 2022.

[25] Z. H. Xia, P. H. Wang, G. G. Dong, *et al.*, "Surrounding prototype loss for radar HRRP open set target recognition," *IEEE Geoscience and Remote Sensing Letters*, vol. 19, article no. 4027305, 2022.

[26] X. J. Ma, K. F. Ji, L. B. Zhang, *et al.*, "SAR target open-set recognition based on joint training of class-specific subdictionary learning," *IEEE Geoscience and Remote Sensing Letters*, vol. 21, article no. 3500805, 2024.

[27] W. T. Li, S. Li, J. Y. Chen, *et al.*, "Spatial distribution learning with multivariate extreme value boundary for radar HRRP open set recognition," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 61, no. 3, pp. 6444–6459, 2025.

[28] Z. H. Xia, P. H. Wang, G. G. Dong, *et al.*, "Radar HRRP open set recognition based on extreme value distribution," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 61, article no. 5102416, 2023.

[29] N. Liao, M. Datcu, Z. H. Zhang, *et al.*, "Analyzing the separability of SAR classification dataset in open set conditions," *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, vol. 14, pp. 7895–7910, 2021.

[30] M. Ezuma, C. K. Anjinappa, M. Funderburk, *et al.*, "Radar cross section based statistical recognition of UAVs at microwave frequencies," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 58, no. 1, pp. 27–46, 2022.

[31] J. Q. Ai, F. F. Wang, Y. X. Mao, *et al.*, "A fine PolSAR terrain classification algorithm using the texture feature fu-

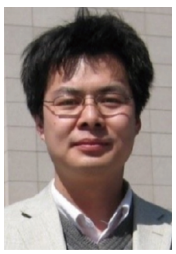
sion-based improved convolutional autoencoder," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 60, article no. 5218714, 2022.
[32] F. Fioranelli, M. Ritchie, and H. Griffiths, "Classification of unarmed/armed personnel using the NetRAD multistatic radar for micro-Doppler and singular value decomposition features," *IEEE Geoscience and Remote Sensing Letters*, vol. 12, no. 9, pp. 1933–1937, 2015.
[33] O. Chapelle and A. Zien, "Semi-supervised classification by low density separation," in *Proceedings of the 10th International Workshop on Artificial Intelligence and Statistics*, Bridgetown, Barbados, pp. 57–64, 2005.
[34] P. D. Grünwald, *The Minimum Description Length Principle*. MIT Press, Cambridge, MA, USA, 2007.
[35] C. M. Bishop, *Pattern Recognition and Machine Learning*. Springer, New York, NY, USA, 2006.
[36] R. A. Fisher and L. H. C. Tippett, "Limiting forms of the frequency distribution of the largest or smallest member of a sample," *Mathematical Proceedings of the Cambridge Philosophical Society*, vol. 24, no. 2, pp. 180–190, 1928.
[37] S. Coles, *An Introduction to Statistical Modeling of Extreme Values*. Springer, London, UK, 2001.
[38] B. Gnedenko, "Sur la distribution limite du terme maximum d'une serie aleatoire," *The Annals of Mathematics*, vol. 44, no. 3, pp. 423–453, 1943.
[39] X. L. Chen, W. Yuan, X. L. Du, *et al.*, "Multi-band multi-angle FMCW radar low-slow-small target detection dataset (LSS-FMCWR-2.0) and feature fusion classification methods," *Journal of Radars*, vol. 14, no. 5, pp. 1276–1293, 2025.
[40] L. Dinh, J. Sohl-Dickstein, and S. Bengio, "Density estimation using real NVP," in *Proceedings of the 5th International Conference on Learning Representations*, Toulon, France, 2017.
[41] L. van der Maaten and G. Hinton, "Visualizing data using t-SNE," *Journal of Machine Learning Research*, vol. 9, no. 86, pp. 2579–2605, 2008.
[42] J. Nair, A. Wierman, and B. Zwart, *The Fundamentals of Heavy Tails: Properties, Emergence, and Estimation*. Cambridge University Press, Cambridge, UK, 2022.
[43] B. Schölkopf, R. Williamson, A. Smola, *et al.*, "Support vector method for novelty detection," in *Proceedings of the 13th International Conference on Neural Information Processing Systems*, Denver, CO, USA, pp. 582–588, 1999.
[44] G. Y. Chen, P. X. Peng, X. Q. Wang, *et al.*, "Adversarial reciprocal points learning for open set recognition," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 44, no. 11, pp. 8065–8081, 2022.
[45] L. Neal, M. Olson, X. Fern, *et al.*, "Open set learning with counterfactual images," in *Proceedings of the 15th European Conference on Computer Vision (ECCV)*, Munich, Germany, pp. 620–635, 2018.
[46] X. Sun, Z. N. Yang, C. Zhang, *et al.*, "Conditional Gaussian distribution learning for open set recognition," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, Seattle, WA, USA, pp. 13477–13486, 2020.
[47] K. Lee, K. Lee, H. Lee, *et al.*, "A simple unified framework for detecting out-of-distribution samples and adversarial attacks," in *Proceedings of the 32nd International Conference on Neural Information Processing Systems*, Montréal, Canada, pp. 7167–7177, 2018.
[48] W. T. Li, S. Li, B. Tian, *et al.*, "Regularization-based region learning for radar HRRP open set recognition," *IEEE Transactions on Instrumentation and Measurement*, vol. 74, article no. 2512615, 2025.
[49] C. X. Geng, S. J. Huang, and S. C. Chen, "Recent advances in open set recognition: A survey," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 43, no. 10, pp. 3614–3631, 2021.
[50] S. Y. Liang, Y. X. Li, and R. Srikant, "Enhancing the reliability of out-of-distribution image detection in neural networks," in *Proceedings of the 6th International Conference on Learning Representations*, Vancouver, BC, Canada, 2018.
[51] B. R. Mahafza, "Radar clutter," in *Radar Systems Analysis and Design Using MATLAB*, B. R. Mahafza, Ed. Chapman and Hall/CRC, New York, NY, USA, 2013.



Guidong He received the B.S. degree in electrical engineering from Beihang University, Beijing, China, in 2021. He is currently pursuing the Ph.D. degree in electronics and information engineering with the School of Electronic and Information Engineering, Beihang University, Beijing, China. His research interests include signal processing and target recognition.
(Email: GuidongHe@buaa.edu.cn)



Peng Lei received the B.S. and the Ph.D. degrees in electrical engineering from Beihang University, Beijing, China, in 2006 and 2012, respectively. He is currently an Associate Professor with the School of Electronic and Information Engineering, Beihang University. His research interests include signal processing, especially in time-frequency analysis and spectral estimation, image processing, target recognition, and machine learning in electrical engineering.
(Email: buaaray@gmail.com)



Jun Wang received the B.S. degree in electronic and information engineering from Northwest Polytechnic University, Xi'an, China, in 1995, and the M.S. and the Ph.D. degrees in electronic and information engineering from Beihang University, Beijing, China, in 1998 and 2001, respectively. He is currently a Professor with the School of Electronic and Information Engineering, Beihang University, and also the Dean of Hangzhou Innovation Institute of Beihang University. His research interests include signal processing, digital signal processor/field programmable gate array real-time architecture, and target recognition and tracking.
(Email: wangj203@buaa.edu.cn)



Jiangyou Zhu received the M.S. degree with the School of Electronic and Information Engineering, Beihang University, Beijing, China. She is currently pursuing the Ph.D. degree with the Department of Information Engineering, The Chinese University of Hong Kong, Hong Kong, China. Her research interest focuses on radar signal processing, especially in DOA disambiguation and target recognition.
(Email: zj124@ie.cuhk.edu.hk)